

PROJECT 08

# AI Mock Interview & Assessment Platform

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Product Roadmap: v1.0 — v4.0

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|---------|-------------------|
| Version | 1.0               |
| Date    | September 2026    |
| Scope   | v1.0 through v4.0 |

*A high-level roadmap for squad brainstorming, feature ideation, user journey mapping, and architecture planning.*

# 1. Product Evolution

| v1.0 ----->       | v2.0 ----->       | v3.0 ----->     | v4.0              |
|-------------------|-------------------|-----------------|-------------------|
| Core AI Interview | Dual-Track Coding | Advanced Modes  | Institutional OS  |
| Resume Ingestion  | Agent Coding      | System Design   | Drive Management  |
| Real-time Agent   | Standalone Exam   | Written/Notepad | Cohort Allocation |
| Dynamic Probing   | SEB / Lockdown    | Voice Agent     | Faculty Analytics |
| Rubric Scoring    | Judge0 Sandbox    | Skill Radars    | Longitudinal OS   |

## v1.0 The Intelligent Conversational Core

*Theme: Can the AI conduct a realistic, context-aware, stateful technical & behavioral interview?*

**The Experience.** A candidate uploads their resume, selects a target role, and enters a live interview cockpit.

### Core Capabilities:

- **Resume Intelligence.** Parsing PDFs, identifying key projects, technologies, and verifiable assertions.
- **Stateful Interview Agent.** An autonomous loop (Observe → Reason → Decide → Act) that tracks conversation state and context.
- **Dynamic Probing & Adaptive Difficulty.** The AI challenges vague answers, asks targeted follow-ups on resume claims, and scales question depth dynamically.
- **Evaluation & Session Archive.** Multi-dimensional score breakdown, identified weak areas, and a full transcript history.

### Brainstorming focus

- *UI Flow:* What does the live cockpit look like? How are AI thinking states visualised?
- *AI Prompts:* How does the agent decide when an answer is sufficient vs when to probe deeper?

## v2.0 The Dual-Track Coding Platform

*Theme: Bridging conversational evaluation with executable, multi-language coding infrastructure.*

**The Experience.** Two distinct coding experiences: (1) live coding inside the interview, and (2) standalone formal exams.

### Core Capabilities:

- **Interactive In-Interview Coding Tool.** The AI agent pauses conversation to launch a live coding workspace, runs tests, and reviews Big-O complexity before resuming dialogue.
- **Standalone Coding Assessment Engine.** Scheduled, formal coding examinations with custom problem pools, test duration, and automated test runners.
- **Secure Exam Environment (SEB).** Anti-cheat controls, tab-blur tracking, fullscreen enforcement, and locked-down test windows.
- **Judge0 Sandbox Execution.** Multi-language isolated code runner (Python, Java, C++, JS) with CPU/memory resource limits.

**Brainstorming focus**

- *Tool Triggering*: When and how does the AI decide to invoke the interactive coding workspace?
- *Execution Safety*: How do we handle infinite loops, compile errors, and runtime limits gracefully?

**v3.0 Multimodal Assessments & Competency Intelligence**

*Theme: Broadening technical evaluation modalities and tracking granular skill progress.*

**The Experience.** Beyond standard Q&A and coding: evaluating system architecture, verbal communication, and sub-skill mastery.

**Core Capabilities:**

- **System Design Assessment Tool.** Interactive architecture round: API design, caching, database trade-offs, failure recovery.
- **Written / Notepad Round.** Free-form scratchpad for algorithmic pseudocode, math reasoning, SQL queries.
- **Voice Interview Engine.** Real-time conversational STT and TTS interviewer.
- **Competency Radar & Skill Gaps.** Granular score breakdown (e.g., Java: 84%, System Design: 52%, Problem Solving: 81%).
- **Personalized 30-Day Remediation.** Automated practice problem sets targeting detected weak areas.

**Brainstorming focus**

- *Architecture UI*: What kind of scratchpad or diagramming tool works best for system design?
- *Voice UX*: How to minimize latency and handle speech interruptions naturally?

**v4.0 The Institutional Career Operating System**

*Theme: Scaling the platform for colleges, placement coordinators, batch drives, and longitudinal analytics.*

**The Experience.** College administrators and faculty manage massive recruitment preparation drives and monitor student readiness across semesters.

**Core Capabilities:**

- **Placement Drive & Program Management.** Defining drive conditions, uploading JDs, configuring round rules, curating question pools.
- **Student Cohort Allocation.** Automated routing of batches across tracks (e.g., 100 Candidates → Coding → AI Tech → HR).
- **Faculty & Placement Analytics Dashboard.** Department-wise readiness comparisons, score distributions, audit reports.
- **Longitudinal Progress Tracking.** Score improvements, recurring failure points, trajectory across semesters.

**Brainstorming focus**

- *Admin Portal*: How should faculty filter, export, and inspect transcripts and code submissions?
- *Batch Automation*: How are cohort rules and drive schedules configured intuitively?

| Version | Focus                     | Target Output                                                       |
|---------|---------------------------|---------------------------------------------------------------------|
| v1.0    | Core Conversational AI    | Live AI Mock Interview Cockpit + Resume Claim Intelligence          |
| v2.0    | Dual-Track Coding         | In-Interview Coding Workspace + Standalone SEB Exam Platform        |
| v3.0    | Multimodal & Intelligence | System Design Tool + Voice Agent + Competency Radars + 30-Day Coach |
| v4.0    | Institutional Platform    | College Drive Manager + Cohort Allocation + Faculty Analytics Hub   |